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REVIEW ARTICLE

Ceftaroline fosamil for community-acquired pneumonia and skin and skin structure infections: a systematic review

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Abstract *Background* Ceftaroline is a parentally administered cephalosporin that has an in vitro expanded spectrum of activity compared with other cephalosporins yet data is conflicting regarding its place in therapy. *Aim of the Review* To compare the efficacy and safety of ceftaroline against standard antibiotic regimens for community-acquired pneumonia (CAP) and complicated skin and skin structure infections (cSSSIs). *Method* The databases of MEDLINE, EBSCO, and Embase were searched up to June 2016. Manual review of references was completed and experts in the field were contacted for unpublished data. Randomized controlled trials of ceftaroline in CAP or cSSSI populations were included. Outcomes included clinical cure, mortality, adverse events, serious adverse events, and discontinuation due to adverse events. Meta-analysis was used to pool results for these outcomes. We performed subgroup analyses for gram positive infections in CAP and infections caused by methicillin-resistant Staphylococcus aureus in cSSSIs. Risk of bias was assessed for all studies. *Results* Six trials (three for each indication) were included, each of which had an unclear or high risk of bias in at least one domain. For CAP, ceftaroline was significantly more efficacious in achieving clinical cure than ceftriaxone [risk ratio (RR) 1.11, 95% confidence interval (CI) 1.04–1.19; $I^2 = 47\%$]. For cSSSIs, there was no significant difference in clinical cure between ceftaroline and vancomycin plus aztreonam (RR 1.01, 95% CI 0.97–1.05; $I^2 = 0\%$). No differences were found for overall mortality, serious adverse events, discontinuation due to adverse events, and overall adverse events. *Conclusion* Ceftaroline is a viable therapeutic alternative for patients with CAP and cSSSIs, yet identified risks of bias and poor external validity preclude it from being recommended as a first-line agent.

Keywords Antibiotics · Ceftaroline fosamil · Cephalosporins · Community-acquired pneumonia · Skin infections

Impacts on practice

- This article provides clinicians with greater understanding of the role of ceftaroline for two pneumonia and skin infections.
- Current evidence suggests that ceftaroline could be considered as a therapeutic alternative for community acquired pneumonia and complicated skin and skin structure infections but more evidence is needed until it can be recommended as a first line option.

Introduction

The increasing prevalence of antimicrobial resistance among organisms of common infections is becoming of significant concern for healthcare professionals and healthcare organizations. Infections with resistant organisms can have serious consequences including prolonged hospital stays, increased mortality, longer duration of illnesses, reduced productivity and increased cost. Of particular concern is the emergence of resistance among

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Streptococcus pneumoniae and *Staphylococcus aureus* in infections such as community- acquired pneumonia (CAP) and complicated skin and skin structure infections (cSSSI) [1].

According to the World Health Organization (WHO), lower respiratory tract infections were considered the fourth most common cause of death in the world in 2012 [2]. The mortality rate for hospitalized patients with pneumonia ranges from 4 to 18% and can reach 50% in patients who are in intensive care units [3]. *Streptococcus pneumoniae* remains the predominant causative bacterial pathogen in CAP [4, 5]. Standard antibiotic therapy for CAP is generally selected empirically and the choice depends on many factors such as patient comorbidities and medication cost. Examples of antibiotics recommended by CAP clinical practice guidelines include an antipneumococcal fluoroquinolone or the combination of a beta-lactam plus a macrolide [3]. Nevertheless, the increase in drug-resistant serotypes of *Streptococcus pneumoniae* may still cause a challenge for clinicians to effectively manage pneumonia [6].

cSSSIs are among the most common infections encountered in the healthcare system, and are most commonly caused by *Staphylococcus aureus* gram-positive cocci. The treatment of these infections has become increasingly challenging due to rising rates of methicillin-resistant *Staphylococcus aureus* (MRSA), which is only susceptible to a limited number of antimicrobials including vancomycin [7]. This highlights the needs for antimicrobial agents that are safe and effective against these resistant pathogens.

Ceftaroline is a parentally administered oxyimino cephalosporin with an expanded spectrum of activity in vitro compared with other cephalosporins, including bactericidal activity against drug-resistant *Streptococcus pneumoniae* and MRSA, and non-extended spectrum beta lactamase producing *Klebsiella pneumoniae*, and *Escherichia coli*. Ceftaroline has been considered in the literature as a "fifth generation" cephalosporin. It is administered as the prodrug ceftaroline fosamil which is then converted to the active metabolite, ceftaroline by plasma phosphates. As with other cephalosporins, it acts by binding to Penicillin Binding Proteins (PBP) on bacterial cell walls leading to cell lysis. It is approved by the the US Food and Drug Administration (FDA) and European Medicines Agency (EMA) for the treatment of cSSSIs and CAP [8–10].

Aim of the review

To compare the efficacy and safety of ceftaroline with standard antibiotic regimens for both CAP and cSSSIs.

Method

Literature search

For this systematic review with meta-analysis we searched MEDLINE (PubMed, U.S. National Library of Medicine, National Institutes of Health), EBSCO and Embase database from inception to August 2016 for reports of randomized controlled trials (RCTs) reported in the English language. We combined the following keyword terms: *ceftaroline, ceftaroline fosamil, PPI-0903,* and *TAK-599,* with the subject headings *pneumonia, soft tissue infections, cSSSI (complicated skin and skin structure infections), cSSTI (complicated skin and soft tissue infections)* and *ABSSSI (acute bacterial skin and skin structure infections).*

Additionally, we searched clinicaltrials.gov up to June 2016 for unpublished or ongoing studies, and we manually screened reference lists of relevant articles. We attempted to contact authors of relevant papers (via email) to request additional unpublished data.

Study selection and data extraction

Inclusion criteria for the review were pre-established by the investigator team. We included full reports of RCTs comparing ceftaroline against any active comparator in human patients with CAP or cSSSI. Two reviewers independently screened all studies by title or abstract for eligibility. Discrepancies were resolved through discussion.

Two reviewers (MH and KJW) independently extracted information from each study in an electronic spreadsheet and resolved any discrepancies through discussion and with input from a third reviewer (RT). Both reviewers independently assessed studies for risk of bias as described in the *Cochrane Handbook* [11]. This handbook gives comprehensive instructions on how to assign risk of bias scores according to information given by each article. We rated each study at low, high or unclear risk of bias in the domains of selection, performance, detection, attrition, reporting and other biases. Other biases included points such as sample size, limitations, study design flaw, stopped early due to a data dependent process, or had some other identified problems. Low risk of bias was generally documented when adequate information was given regarding each point and no sources of bias could be identified. Unclear risk of bias was documented when information was lacking to form an opinion. High risk of bias was documented when described procedures or analysis techniques were known to introduce bias, as per the *Cochrane Handbook.*

Primary outcome

The primary outcome was clinical cure as defined for each indication in the original studies. For CAP, clinical cure was defined as total resolution of all signs and symptoms of pneumonia or improvement of signs and symptoms to such an extent that no further antimicrobial therapy was necessary. For cSSSIs, clinical cure was defined as resolution of all signs and symptoms or improvement of signs and symptoms to such an extent that no further antimicrobial therapy was necessary. Outcomes were consistent between studies.

Secondary outcomes

Secondary outcomes included mortality, incidence of adverse events, serious adverse events, and discontinuation due to adverse events. Adverse events were analyzed as a whole, as low numbers precluded analysis of individual events. Other extracted data included funding sources, author associations, and declared conflicts of interest.

Subgroup analyses

For each indication, we performed one pre-defined subgroup analysis of the primary outcome. For CAP, we evaluated whether a documented gram-positive infection impacted the comparative efficacy of ceftaroline. For cSSSIs, we evaluated whether a documented MRSA infection impacted the comparative efficacy of ceftaroline.

Statistical analysis

We performed all statistical analyses using Review Manager Version 5.3. We present dichotomous data as relative risk (RR) with 95% confidence intervals (CIs). We analyzed all randomized patients according to the group to which they were allocated, in accordance with the intention-to-treat (ITT) principle. We assumed that patients lost to follow-up before the end of the trial had not achieved clinical cure.

We assessed for statistical heterogeneity by visual inspection of the forest plot and calculation of the I^2 statistic. We defined $I^2 < 25\%$ as low heterogeneity, 25–50% as moderate heterogeneity, and >50% as severe heterogeneity. When meta-analysis was possible because of acceptable clinical and methodological heterogeneity, we report the fixed-effects model summary estimate for $I^2 < 25\%$ and the random-effects model summary estimate for $I^2 \geq 25\%$. We set the threshold for a positive test for interaction for subgroup analyses at $p \leq 0.10$. We made no adjustments for multiple comparisons.

We intended to assess for publication bias by visual inspection of the funnel plot, but we could not as this test is considered to be of little value in a meta-analysis with fewer than 10 trials [12].

Results

We identified 224 reports, and included six RCTs, three for each indication (Fig. 1). Supplementary Table lists study characteristics. The included trials assessed ceftaroline in diseased patients against comparator agents. Comparator agents included ceftriaxone in CAP studies and vancomycin in addition to aztreonam in cSSSI studies. In one CAP trial, patients in both study groups received macrolide therapy [13]. Generally, study definitions of CAP consisted of radiographically confirmed pneumonia with new or progressive pulmonary infiltrates on chest radiograph or CT scan suggestive for bacterial pneumonia, acute illness less than or equal to seven days with at least three clinical signs or symptoms of pneumonia [13–15]. The diagnosis of cSSSI included patients with deep extensive cellulitis, major abscess, infected wound or abscess, or infected burn that warranted greater than or equal to five days of parenteral antibacterial therapy [16–18].

Risk of bias results are given in Table 1. None of the included trials had low risk of bias in all domains. Five RCTs adequately described sequence generation and allocation concealment [13–17]. whereas it was unclear in the sixth trial [18]. Five trials had a low risk of bias due to blinding [13–17]. All studies provided enough information

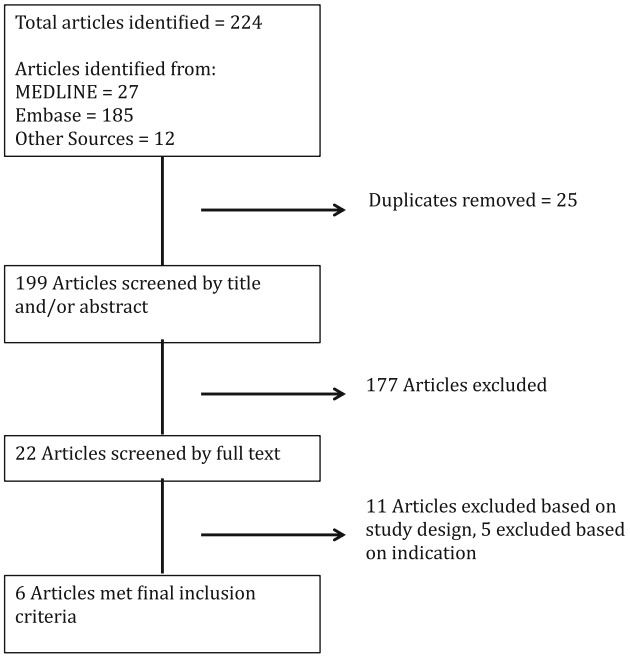

Fig. 1 Flow diagram for study selection and inclusion

Table 1 Risk of bias assessments of included studies

Study	Sequence generation	Allocation concealment	Blinding	Incomplete outcome data	Selective outcome reporting	Other	Explanation
FOCUS 1 [13]	Low	Low	Low	Unclear	Low	Unclear	Lacking details in outcome assessments
FOCUS 2 [14]	Low	Low	Low	Unclear	Low	Unclear	Lacking details in outcome assessments
Zhong 2014 [15]	Low	Low	Low	Low	High	Unclear	High industry involvement
CANVAS 1 [16]	Low	Low	Low	Low	High	Unclear	Lacking details in outcome assessments
CANVAS 2 [17]	Low	Low	Low	Low	High	Unclear	Imbalance in baseline characteristics and lacking details in outcome assessments
Talbot 2007 [18]	Unclear	Unclear	High	Low	Low	High	Small sample size and lack of consistency in standard therapy between patients

for ITT analyses and did not have a high risk of attrition bias [13–18]. Three studies were ranked high for selective outcome reporting due to lack of reporting follow-up results [13–15] and all studies were given a high or unclear risk of other biases [13–18]. Other bias rankings were largely due to a lack of reporting in outcome assessments for each study. The unclear ranking was given, as this information was not provided.

Primary outcome

Clinical cure

All six trials provided sufficient data on clinical cure, as defined in Supplementary Table. In CAP, ceftaroline was statistically significantly more efficacious in achieving clinical cure compared to ceftriaxone using a random-effects model [risk ratio (RR) 1.11, 95% confidence interval (CI) 1.04–1.19] (Fig. 2). There was moderate statistical heterogeneity ($I^2 = 47\%$), primarily explained by the trial by Zhong et al. [15].

In cSSSI, there was no statistically significant difference in clinical cure between ceftaroline and vancomycin plus aztreonam (RR 1.01, 95% CI 0.97–1.05, $I^2 = 0\%$). For both indications, fixed- and random-effects models did not differ.

All three CAP trials reported clinical cure in subgroups of patients with and without Gram-positive bacteria on culture. There was no difference in treatment effect between subgroups (test for interaction $p = 0.12$).

Similarly, all three cSSSI trials reported no difference in clinical cure outcomes between the subgroups of patients with and without MRSA (test for interaction $p = 0.52$).

Secondary outcomes

Mortality

All six included trials reported on all-cause mortality over study follow-up. There was no statistically significant difference in mortality between ceftaroline and control using a fixed-effects model (RR 1.30, 95% CI 0.69–2.45, $I^2 = 0\%$) (Fig. 3).

Adverse events: serious adverse events, discontinuations and overall

All six included trials reported adverse events rates. There was no statistically significant difference between ceftaroline and control in any of serious adverse events (RR 0.99, 95% CI 0.79–1.25, $I^2 = 0\%$), discontinuation due to adverse events (RR 0.83, 95% CI 0.57–1.21, $I^2 = 0\%$), or overall adverse events (RR 0.98, 95% CI 0.90–1.06, $I^2 = 7\%$) using a fixed-effects model (Fig. 4).

Discussion

In this systematic review with meta-analysis, we found that ceftaroline was significantly more efficacious at achieving clinical cure for patients with CAP compared to ceftriaxone. We found no significant difference between ceftaroline and comparators for cSSSIs. However, results do not overlap the 10% absolute risk difference non-inferiority margin. No difference in subgroups was found for those with or without documented MRSA cSSSIs.

For clinical cure of CAP, one trial had a greater effect of ceftaroline compared to the others [15]. This trial was completed solely in Asian centers and differences in care standards or supportive care procedures may have influenced this finding. It is also possible that these centers may have higher rates of resistance to control agents [13, 14]. Reasons for this discrepancy are unclear and therefore may have biased results in favor of ceftaroline.

For cSSSIs, the results must be interpreted cautiously. The included trials have poor generalizability owing to design choices that deviate from real-world practice. Study protocols forbade oral step down and outpatient therapy, even in clinically stable patients, which counters current

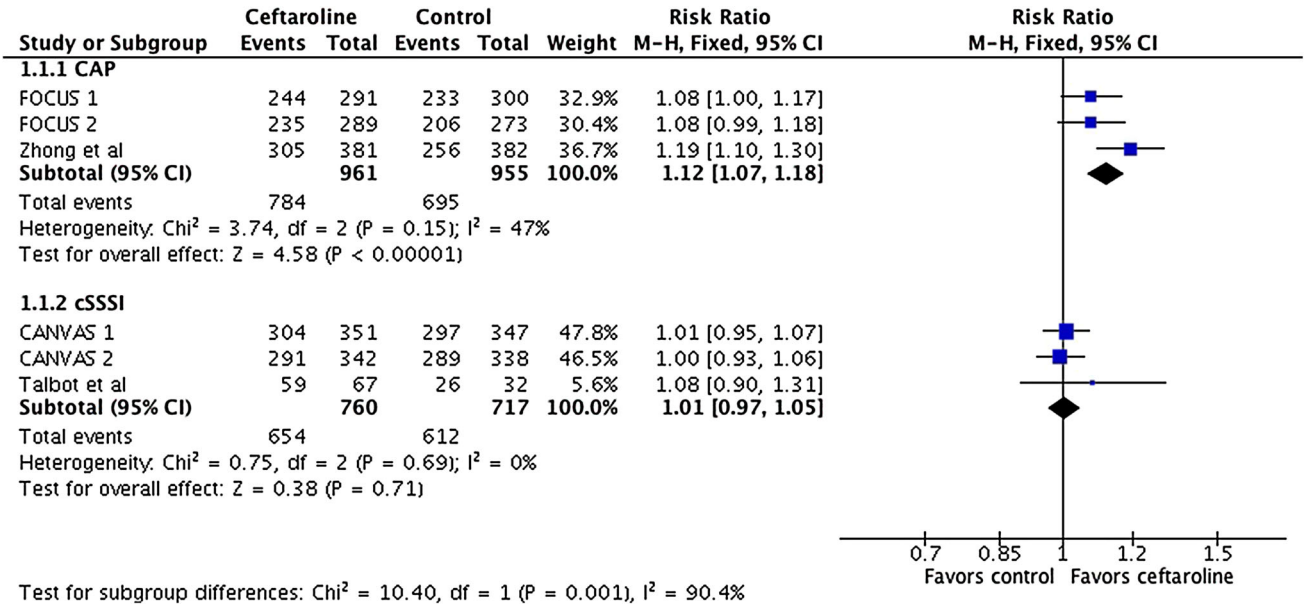

Fig. 2 Effect of ceftaroline on clinical cure for community-acquired pneumonia and chronic skin and soft tissue infections

best practices and antimicrobial stewardship principles. Second, the comparator agents do not represent first-line therapies recommended by infectious diseases guidelines. For example, the widely-used Infectious Diseases Society of America (IDSA) guidelines recommend monotherapy with an antibiotic with activity against gram-positive bacteria for cSSSIs, yet all ceftaroline cSSSI studies used a combination of vancomycin and aztreonam as the comparator [19].

We identified no statistically significant difference in mortality for either indication, which may have resulted from a lack of power owing to a low number of events. Alternatively, this may have been due to exclusion of patients admitted to the ICU, where most CAP mortality occurs. While mortality is largely unexpected for cSSSIs, one would expect mortality to be higher for hospitalized patients with CAP. However, the majority of patients enrolled were of PORT score III, for whom risk of death is

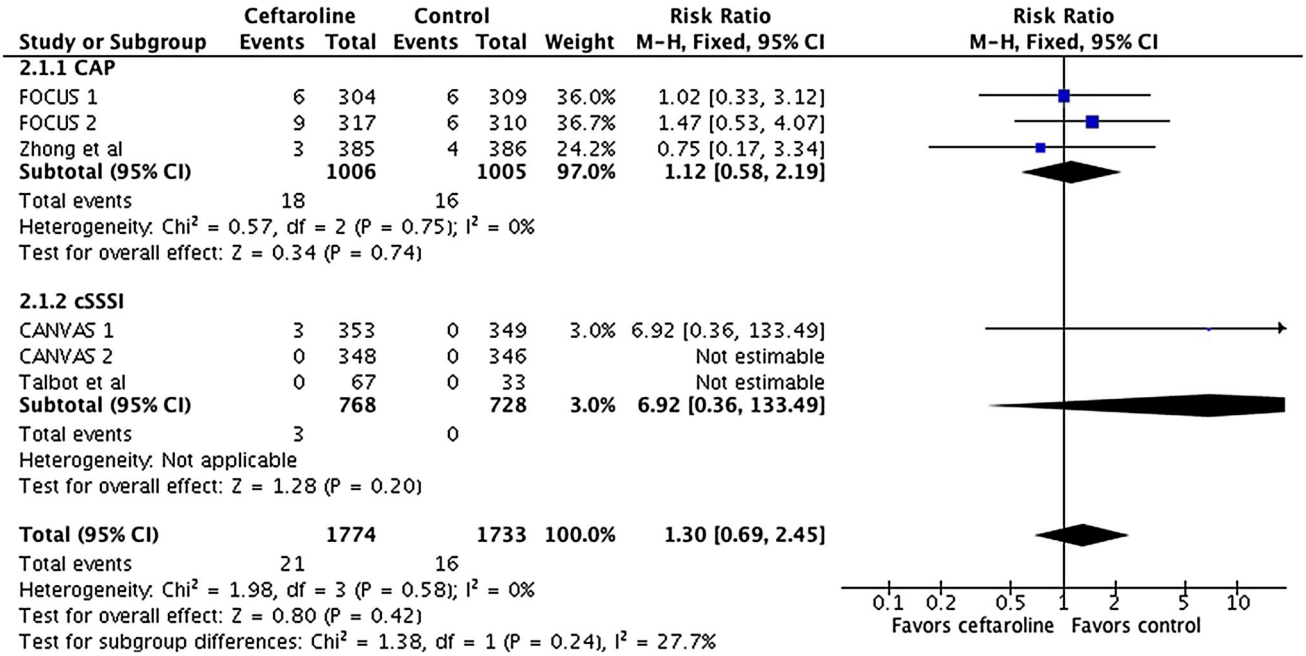

Fig. 3 Effect of ceftaroline on mortality for community-acquired pneumonia and chronic skin and soft tissue infections

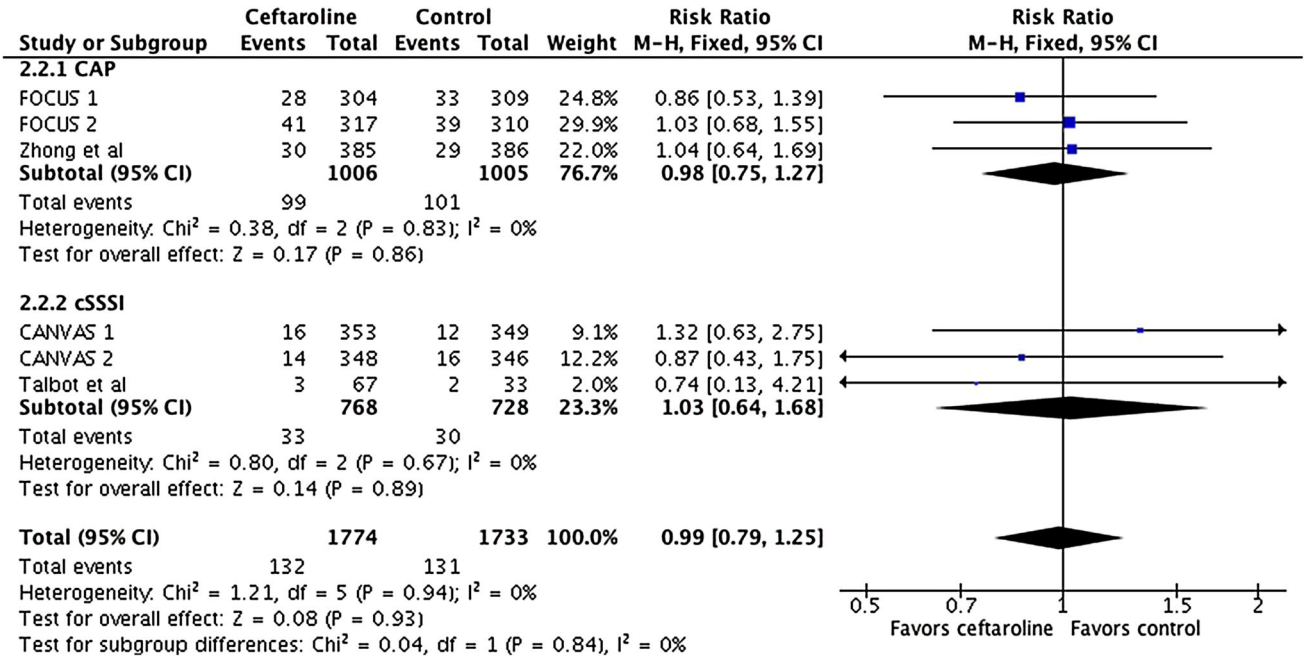

Fig. 4 Effect of ceftaroline on serious adverse events in community-acquired pneumonia and chronic skin and soft tissue infections

generally low and use of intravenous antibiotics is typically unnecessary [20]. Thus, one likely explanation for the lack of difference in mortality may simply be insufficient power.

No major findings were noted for safety outcomes, with ceftaroline having similar adverse events and discontinuation rates compared to controls.

The major strength of our review is the risk of bias assessment that other reviews on the topic have neglected to include. The included studies had high or unclear risk of bias in at least one domain that may have influenced results and authors' original conclusions. First, ITT was not used for the primary analysis and conclusions were largely made on only the clinically evaluable populations. Fortunately, reports included sufficient information for us to extract ITT numbers and determine relative risks based on this more conservative and appropriate measure. Second, definitions, procedures, and results of recurrence and relapse were inconsistently reported between studies, demonstrating possible selective outcome reporting bias. Third, the risk of industry influence (studies were industry funded or authors employed by industry companies) was high throughout all studies and many conclusions made by authors were unjustifyingly in favor of ceftaroline based on the results. For example, Zhong, et al. [15] concluded that 'ceftaroline should be used as a comparator in future clinical trials designed to support the approval of new antibiotics for the treatment of community-acquired pneumonia'. This statement is unfounded until ceftaroline's role is established within CAP and it becomes a gold standard comparator agent. These bias concerns call for more conservative conclusions regarding ceftaroline's place in therapy, as identified below.

Conclusion

Although our results suggest that ceftaroline is a safe and effective therapeutic alternative for patients with CAP and cSSSIs, its place in therapy is unknown for both indications. As a broad-spectrum agent, use should be reserved until efficacy and safety is established for patients presenting with resistant infections, such as MRSA. Early, widespread use may promote resistance to cephalosporins and other beta-lactams. Therefore, we recommend that ceftaroline be reserved as a therapeutic alternative for those patients unable to take or tolerate standard treatment at this time and that clinicians continue to employ good antimicrobial stewardship principles, including switching from ceftaroline to a more targeted antibiotic once bacterial culture and susceptibilities become available.

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Conflicts of interest All authors report no conflicts of interest.

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